

EKS Provisioning using Pulumi

- We will provision the EKS using Pulumi as an Infrastructure as Code.
 - We will deploy it in a custom Virtual Private Cloud for isolation.
 - We will also deploy RDS MySQL Instance to store the relational data and connect it to EKS.
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Prerequisites

1. An AWS account with an IAM user having sufficient permissions.
 2. AWS CLI installed and configured with the IAM user.
 3. Pulumi Installed.
 4. Kubectl Installed.
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Write Pulumi Configuration files

First, we will initiate and edit Pulumi configuration files for AWS resources using predefined Pulumi Library available on the internet.

Steps

1. Create a Pulumi Project directory.
2. Open the PowerShell.
3. Change the directory to the above-created Pulumi Project.
4. Run the `pulumi new aws-python` command to initialize the *pulumi*.
5. Provide the appropriate values to prompts such as *project-name*, *project-description*, *stack-name*, *toolchain*, *region-name*, etc.
6. This will generate some Pulumi files in this directory.
7. Now we will install predefined Pulumi modules.
8. Activate the `venv` by running `venv\Scripts\activate`.
9. Run `pip install git+https://github.com/inflection-sahil/pulumi.git` to install the modules.
10. Deactivate the `venv` by running `deactivate`.
11. Now open the directory in the preferred IDE.
12. Create *commons* folder
13. Inside the folder create *init.py* file.
14. Import the following in the *init.py* file:
 - `from inflection_zone_pulumi.modules.aws.vpc import vpc`
 - `from inflection_zone_pulumi.modules.aws.rds import rds`
 - `from inflection_zone_pulumi.modules.aws.eks import eks`
15. Click [code](#) for reference.
16. Definition of *init.py* is complete.
17. Now create the *values.py* file in the root folder of the above-created project directory.
18. Define the following values:

- vpc_properties
- rds_properties
- bastion_properties
- eks_properties

19. Click [code](#) for reference.

20. The definition of *values.py* is complete.

21. Now navigate to the **main.py** file present in the root folder of the above-created project directory.

22. Clear the sample code if present.

23. Import the following:

- from commons import vpc, rds, eks
- values

24. Define the following objects and pass the values as an argument:

- VPC
- RDS
- EKS

25. Click [code](#) for reference.

26. Definition of **main.py** is complete.

Provisioning the Infrastructure

Now we will provision the infrastructure by applying the above-created configuration files.

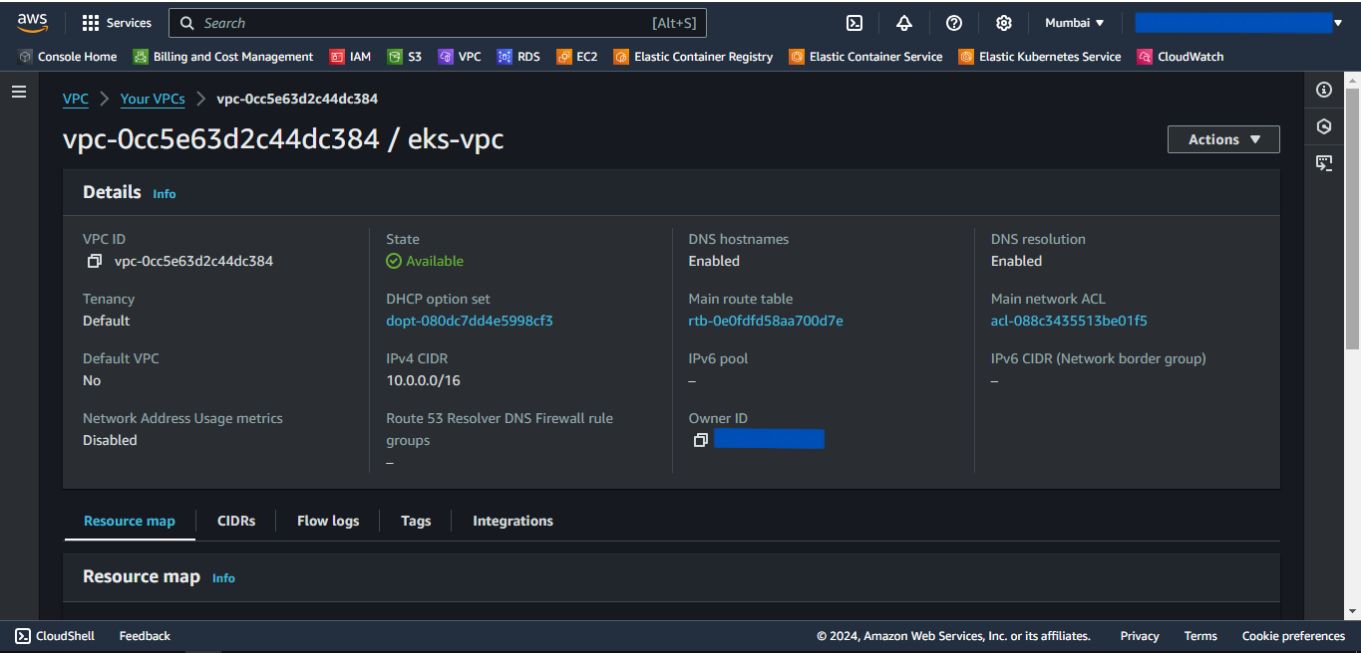
Ensure AWS CLI is configured with appropriate IAM user credentials and enough permissions.

Steps:

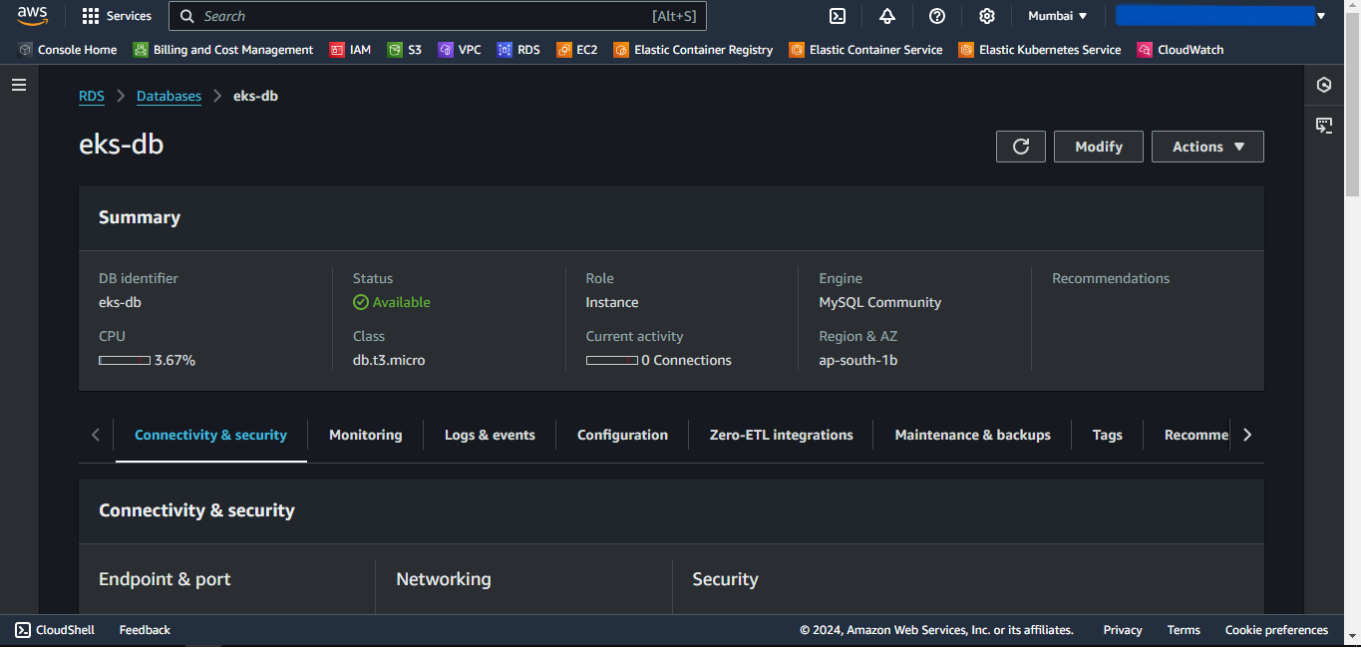
1. Open the PowerShell.
 2. Change the directory to the above-created Pulumi Project.
 3. Run the **pulumi up** command and if prompted, select **yes** to provision the infrastructure onto the AWS Cloud.
 4. Head to the AWS Console, and verify the created resources.
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Screenshots of Provisioned Infrastructure

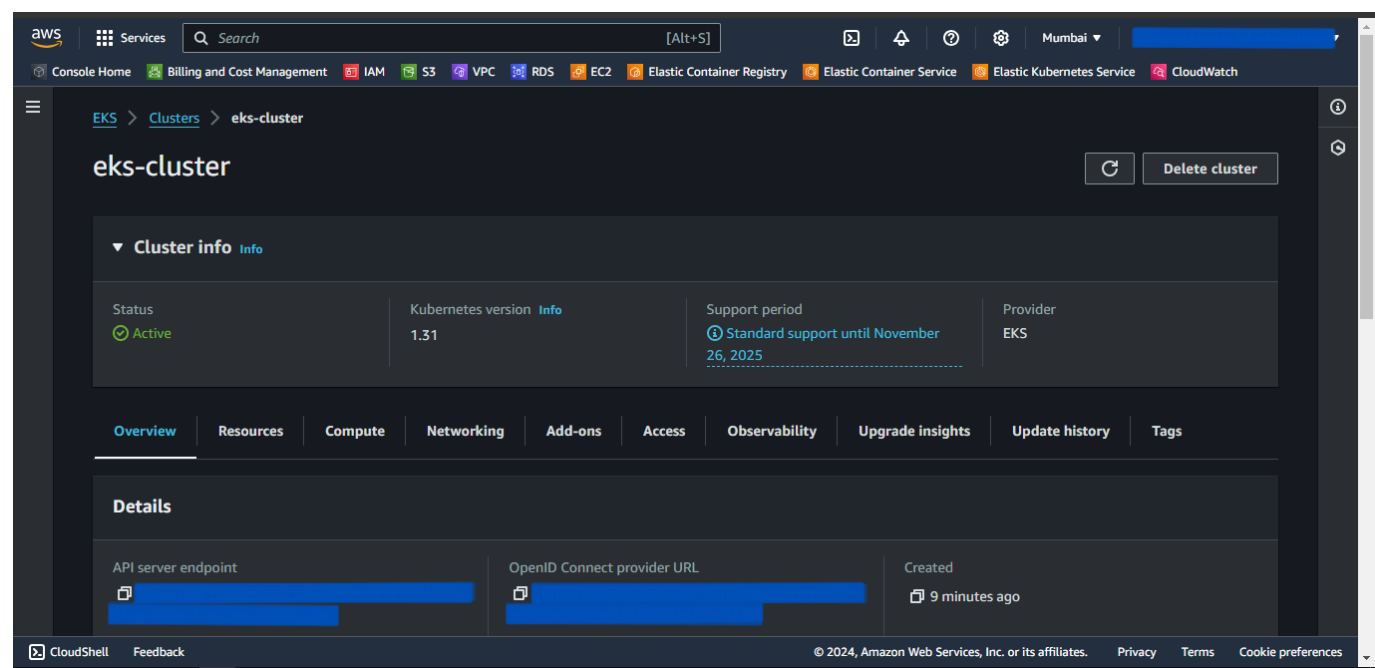
VPC Image



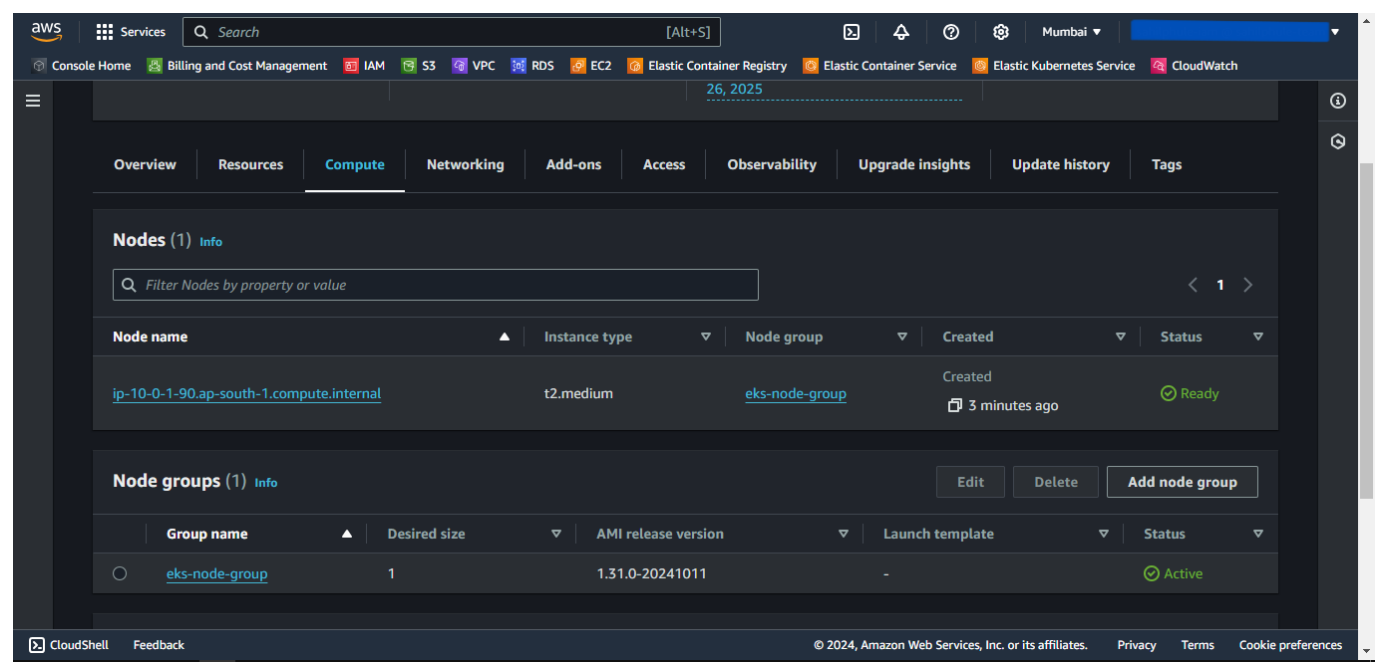
RDS Image



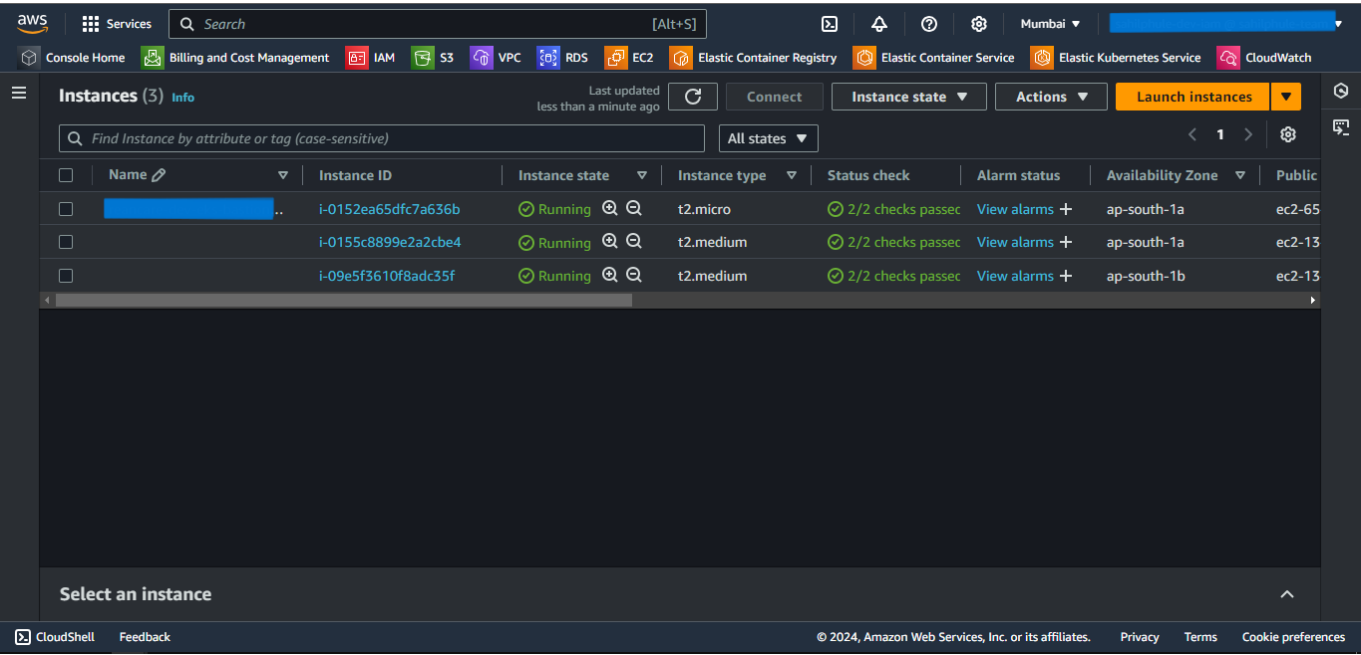
EKS Cluster Image



EKS Node Group Image



EKS Nodes Image



Connect to EKS Cluster from Powershell

Steps

- 1. Open a new Powershell window.
- 2. Run the following command to configure local kubectl with eks cluster

```
aws eks --region <region-name> update-kubeconfig --name <cluster-name>
```

Substitute *<region-name>* and *<cluster-name>* with the values defined in the above-created *values.py* file.

- 3. Now, apply the Kubernetes manifest files for the application.
- 4. To list them all, run `kubectl get all`.

Connection to the RDS database through Bastion Host using MySQL Workbench

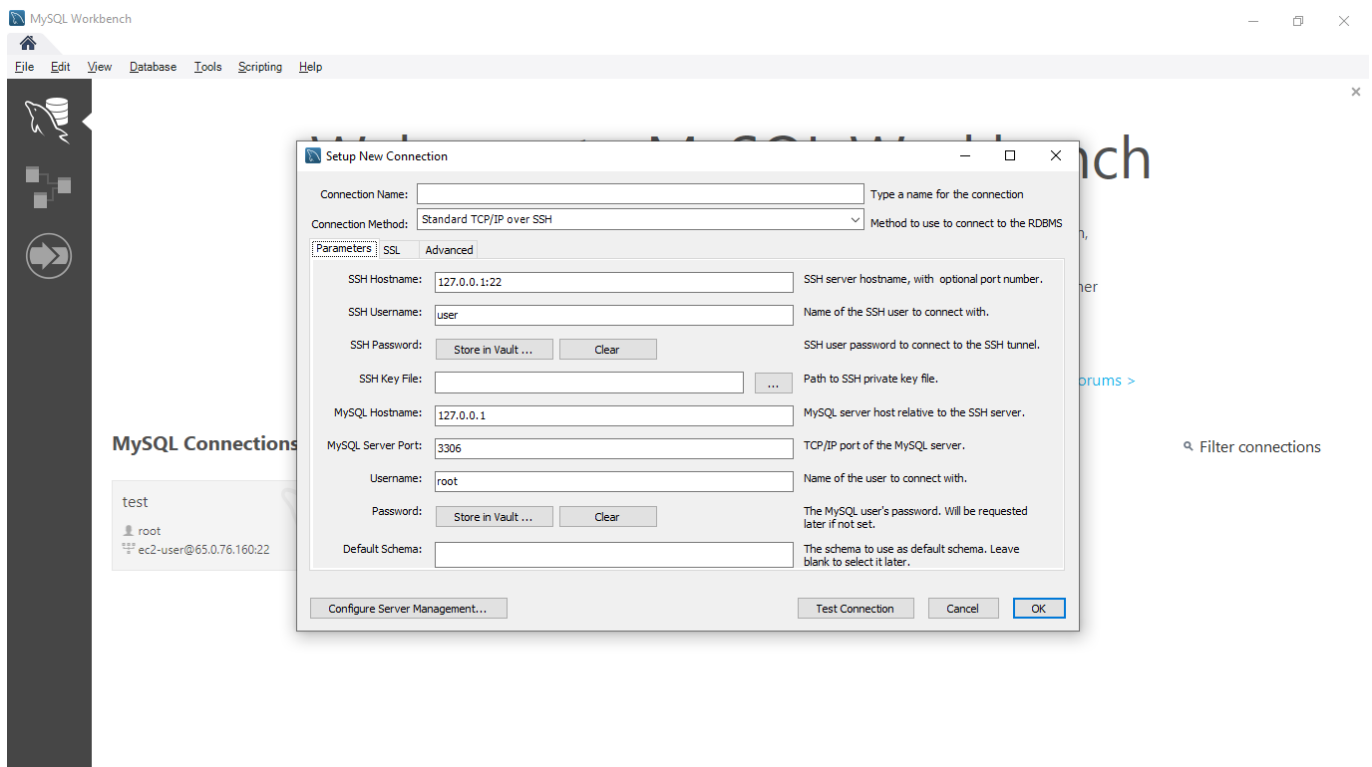
Now, we will use MySQL Workbench to connect and access the MySQL RDS Database through above created Bastion Host.

- 1. Open MySQL Workbench.
- 2. Click Add Connection.
- 3. Select connection method as **Standard TCP/IP over SSH**.

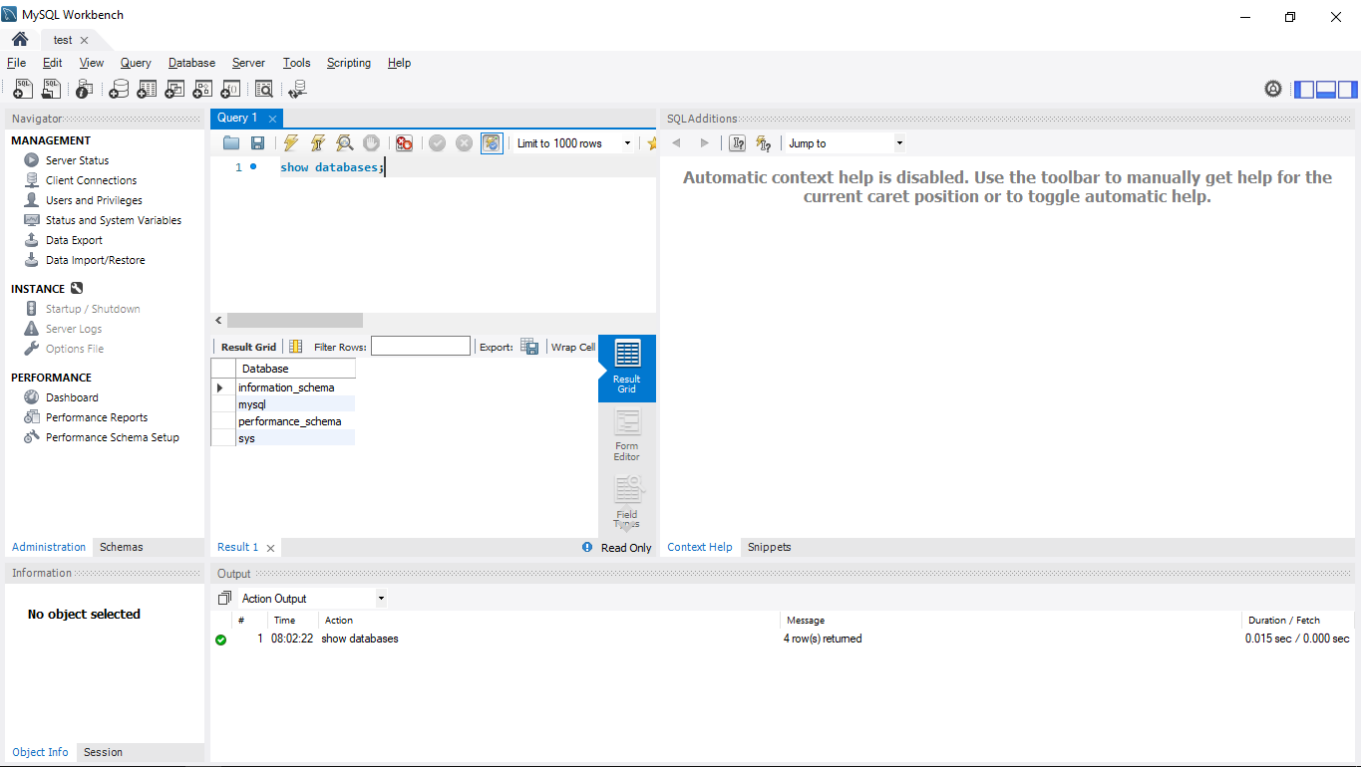
4. In SSH Hostname, enter *bastion-host-ip:22* where *bastion-host-ip* is received from **pulumi stack output bastion-host-ip** command.
5. In SSH Username, enter *ec2-user*.
6. In SSH Key File, select *bastion-key.pem* file passed in above *values.py* file from your local computer.
7. In MySQL Hostname, enter *DB_HOST* where *DB_HOST* is received from **pulumi stack output DB_HOST**.
8. In the Password section, select *Store in Vault*, and enter the password passed in above-created *values.py* file.
9. Click *OK* and open the connection.
10. Now you can run MySQL commands to access databases and verify the successful connection of *eks-nodes*.

Screenshots of MySQL Workbench

Connection Page



Commands Page



Destroy the provisioned infrastructure

Lastly, we will destroy the above-created resources.

Steps

1. First, delete all the Kubernetes Deployments.
2. To destroy infrastructure, open the Powershell Window and change the directory to the above-created Pulumi Project using the `cd` command.
3. Run `pulumi destroy` & if prompted, select `yes`.
4. Infrastructure will be destroyed.